SIMULATION EXPERIMENT – 6

INDUSTRIAL HVAC CONTROL FIRMWARE: ADC TEMPERATURE ACQUISITION AND TIMER-INTERRUPT PWM FAN CONTROL

➔ **PROGRAM:**

```
#define TEMP A0

#define FAN  8


volatile byte pwm = 0;

volatile byte count = 0;


ISR(TIMER1_COMPA_vect)

{

 count++;


 if (count >= 100)

  count = 0;


 digitalWrite(FAN, count < pwm);

}


void setup()

{

 Serial.begin(9600);

 pinMode(FAN, OUTPUT);


 TCCR1A = 0;

 TCCR1B = (1 << WGM12) | (1 << CS11);
```

```
  OCR1A = 199;

  TIMSK1 = (1 << OCIE1A);

  sei();
}

void loop()
{
  int adc = analogRead(TEMP);

  float temp = adc * 500.0 / 1023.0;

  if (temp < 25)
    pwm = 0;
  else if (temp < 30)
    pwm = 30;
  else if (temp < 35)
    pwm = 60;
  else
    pwm = 90;

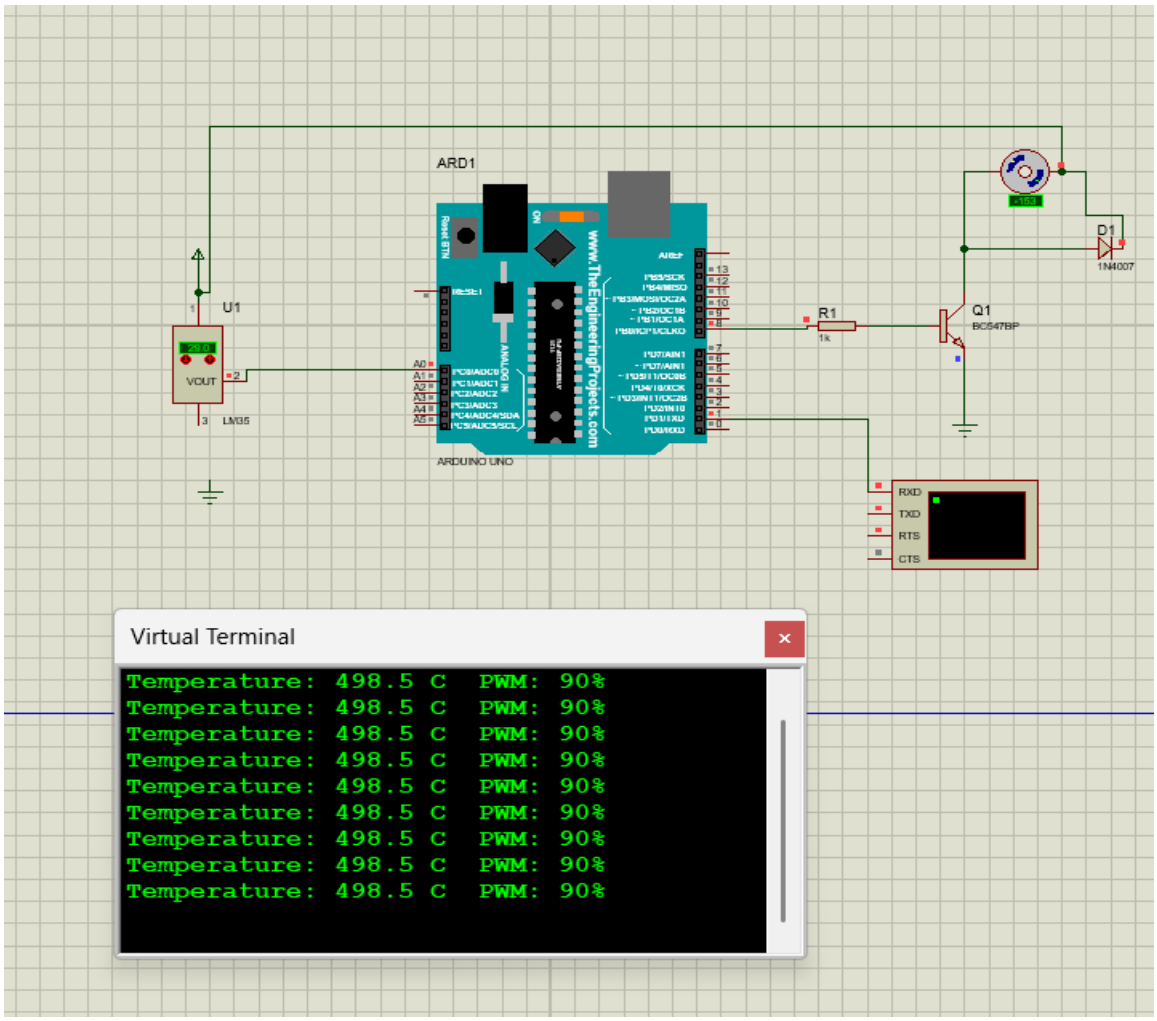
  Serial.print("Temperature: ");
  Serial.print(temp, 1);

  Serial.print(" C  PWM: ");
  Serial.print(pwm);
```

```
 Serial.println("%");


 delay(1000);
}
```

Output



Reflection

Through this experiment, I learned how to acquire temperature data using an ADC and control fan speed using PWM generated by timer interrupts. I understood how temperature readings can be used to adjust the fan speed automatically in an industrial HVAC system. This experiment helped me gain practical knowledge of ADC, timers, interrupts, and PWM control, which are important for efficient temperature regulation and energy saving.